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Quiz

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Main notions learned in this chapter

4/6 points (graded)

For a given Universal Turing Machine U , there exists a specific Universal Turing Machine V which can compute the complexity C_U relative to machine U .

☒ True

☐ False
✓



Answer

Incorrect: No! Complexity is not computable.

Prefix complexity is defined on Turing machines on which all programs terminate.

☐ True

☒ False



Answer

Correct:

No. Prefix complexity is defined on Universal Turing machines that only accept self-delimited programs.

The programs for which a prefix Turing machine halts corresponds to the code-words of a prefix code.

☒ True

☐ False



Answer

Correct:

Yes. Both satisfy the property that one string cannot be the prefix of any other string.

Can a decimal sequence of length 1000 be random if it appears in the decimal development of π ?

☐ Yes
✓

☒ No



Answer

Incorrect:

Yes. π 's decimals as a whole are simple and thus are not random. However, a chunk of π 's decimals is as complex as the location where it starts. So if that location is larger than (2^{1000}) , the sequence can genuinely be regarded as random.

Chaitin's version of Gödel's (first) incompleteness theorem proves that the complexity of some integers is so large that they cannot be proven to be that complex.

☒ True

☐ False



Answer

Correct:

Yes. Above a certain level of complexity m , it is impossible to prove that an integer n is more complex than m . The value of m depends on the mathematical theory used.

Chaitin's number Omega is the probability that programs will stop when executed. Since proofs are programs, we may imagine that Omega can be used to decide whether Goldbach's conjecture is true or not, by looking at a specific location in its development. If so, would it constitute a proof of the conjecture?

☐ Yes

☒ No



Answer

Correct:

No. Since Omega is uncomputable, a location in Omega's development brings no information.

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You have used 1 of 1 attempt

i Answers are displayed within the problem

Problem 1

2.0/2.0 points (graded)

π is believed to be a normal number, which means that its expansion contains all finite numbers. We consider the location L of the first occurrence of a given number n in π 's decimal expansion. What is at most the complexity of L in bits?

☒ $\log_2(n) + O(1)$

☐ $\log_2(10^n) + O(1)$

☐ $\log_2(n^{10}) + O(1)$

☐ $\log_2(\log_2(L)) + \log_2(n) + O(1)$

☐ $\log_2(\log_2(L)) + \log_2(10^n) + O(1)$

☐ $\log_2(\log_2(L)) + \log_2(n^{10}) + O(1)$



Answer

Correct:

To find n 's location L , we just need to know n . Then we determine L by using a small program that scans π 's expansion. So $\log_2(n) + O(1)$ is the right answer.

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Problem 2

2.0/2.0 points (graded)

What is the best upper bound for the set $\{C(x); 0 \leq x \leq 1000\}$, where the x are rational numbers?

☐ $\log_2(1000)$

☐ $2 \times \log_2(1000)$

☒ No upper bound



Answer

Correct:

There is no upper bound. There are infinitely many rationals smaller than 1000 with arbitrary complexity.

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Problem 3

0/2 points (graded)

Can randomness be considered an objective notion?

☐ No, it depends on the computing machine.

☐ Yes for finite sequences; No for infinite sequences.

☐ No for finite sequences; Yes for infinite sequences.



☒ Yes for both finite and infinite sequences.



Answer

Incorrect:

Some finite sequences that are incompressible on certain machines can be compressed down to one bit on another machine that holds the sequence in its memory. So Randomness is not objective for finite sequences. It is, however, objective for infinite sequences: if one machine can compress the sequence, any other universal Turing machine can compress it by emulating that machine.

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